

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. BPT-604: Atomic Physics and Lasers

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer all questions. Symbols have their usual meanings.

Question 1. Answer any **five** of the following: [5 × 4 = 20]

- (a) What is the relative population of the two states in a ruby laser that produces a light beam of wavelength 6943 Å at 27°C?
- (b) Why is the discharge tube in a He-Ne laser system taken to have a narrow cross-section?
- (c) The effect of nuclear mass on the Rydberg constant is to change it by about 1 part in 2000. Justify this statement.
- (d) Which experiment provided evidence for the existence of electron spin, and how?
- (e) Explain the meaning of space quantization.
- (f) On introducing the concept of the relativistic electron in the elliptic Sommerfeld orbit treatment, the splitting of energy levels is the same as obtained by Dirac's theory. Do these two explain the observed structure completely? If not, why?
- (g) Work out the ground state of an atom having atomic number 39.

Question 2.

- (a) Describe the construction and working of an Nd-YAG laser. [12.5]

OR

- (a) With the help of a suitable diagram, describe the construction and working of a He-Ne laser. Which one of the He and Ne atoms is responsible for the LASER action and what is the role of the other? [12.5]

Question 3.

- (a) What is the difference between spontaneous and stimulated emissions? Establish the relation between the Einstein coefficients A and B . [12.5]

OR

- (a) An emitting atom is placed in a magnetic field. If the field intensity is increased from zero to a low value (a few Gauss), the singlet lines split into three components. Explain the origin of this triplet. [12.5]

Question 4.

- (a) What do you understand by Landé's interval rule and what are normal and inverted multiplets? Explain with suitable examples. [7.5]
- (b) Find the ground state of an atom with two equivalent p electrons. [5]
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Question 5.

- (a) Describe the Franck-Hertz experiment and its outcome. [4.5]
- (b) Discuss briefly the condition of phase matching in Second Harmonic Generation (SHG). [4]
- (c) Derive terms for $5d6p$ electrons under $j - j$ coupling scheme. [4]